Report on the Design of the Reservoir and Nozzle System for the Automated Fertilizer Spraying Drone

---

1. Introduction

1.1 Project Overview

The rise of automation in agriculture has provided substantial benefits in terms of efficiency, sustainability, and precision. One of the most promising applications of this technology is the **automated fertilizer spraying drone**, which allows for precise and efficient distribution of fertilizers over large agricultural fields. This technology is aimed at reducing human labour, minimizing fertilizer wastage, and ensuring that fertilizers are applied more accurately and uniformly, which ultimately improves crop yield and environmental sustainability.

The drone in question is designed to fly autonomously over fields, equipped with a fertilizer reservoir and nozzle system that distributes fertilizer based on pre-programmed flight paths, field characteristics, and GPS data. The drone is equipped with sensors and advanced control systems that enable it to adjust its spraying rate, altitude, and speed dynamically to optimize fertilizer usage and crop health.

As part of the design team, **Hari** and I were specifically tasked with designing the **reservoir** (the component that holds the fertilizer) and the **nozzle system** (the component that releases the fertilizer in a controlled manner). These components are crucial to the drone's performance, as they directly impact the efficiency, accuracy, and functionality of the entire system.

This report aims to provide a comprehensive overview of the design, development, and optimization of these two components. We will discuss the **initial conceptualization**, **design requirements**, **material selection**, **mounting considerations**, **testing and validation**, and **final results**. Our objective is to demonstrate how we addressed various engineering challenges in designing the reservoir and nozzle system to meet the project's technical and operational goals.

1.2 Importance of the Design Phase

The **design phase** of any engineering project is critical because it establishes the foundation upon which all subsequent work—such as prototyping, testing, manufacturing, and final production—is built. The design of the **reservoir** and **nozzle system** for an **automated fertilizer spraying drone** is no different. A well-thought-out design not only ensures that the system will meet technical, environmental, and operational requirements but also minimizes the risk of errors during production and improves overall system performance. Below, we dive deeper into the **importance of the design phase**, highlighting its significance in various aspects of the project.

1.2.1 Alignment with Project Objectives and Requirements

The design phase is the stage where the overall objectives and technical requirements of the project are translated into tangible, executable plans. For a project like the **automated fertilizer spraying drone**, where precision and efficiency are paramount, it is essential that the design meets the following criteria:

- **Payload capacity**: The drone must be able to carry the reservoir and nozzle system without compromising its ability to fly efficiently. Ensuring that the components do not exceed the drone's payload limits is critical for maintaining flight stability and operational efficiency.
- **Spray accuracy and uniformity**: The nozzle system must distribute the fertilizer in a precise and controlled manner. Without a thorough design phase, achieving uniform coverage of large fields would be impossible, resulting in wasted fertilizer, crop damage, or inefficiency.
- **Chemical resistance**: The materials used for both the reservoir and nozzle must withstand the corrosive nature of the fertilizers being sprayed. An incorrect material choice could lead to early system failures, requiring costly repairs or replacements, and potentially ruining crop yields.

The design phase helps align all components of the drone with the project's technical and operational goals, ensuring that the system works as intended and that all components are integrated seamlessly. Without careful planning during the design phase, achieving these objectives becomes a matter of chance, which could result in failure.

1.2.2 Identifying and Mitigating Potential Risks

Every engineering project is subject to risks—whether they are related to performance, safety, or cost. The design phase is the first opportunity to **identify** and **mitigate** these risks before they have a chance to affect the project. In the case of the fertilizer spraying drone, several risks must be considered:

- **Structural Integrity**: The reservoir and nozzle system must be designed to withstand external forces, such as wind during flight, as well as the internal pressure from the fertilizer fluid. The risk of a **ruptured reservoir** or a **broken nozzle** is one of the primary safety concerns. A poor design can compromise the strength and durability of these components, leading to catastrophic failures during operation.
- **Aerodynamic Impact**: The addition of a heavy or improperly placed reservoir could affect the drone's **center of gravity** or **aerodynamic properties**, leading to unstable flight or reduced performance. If the design phase does not account for the weight distribution, flight stability could be compromised, making the drone harder to control or reducing flight times.
- **Flow Consistency**: One of the main challenges of the nozzle system is ensuring consistent flow rates despite variations in pressure as the fertilizer level decreases during flight. Inadequate flow control can lead to uneven fertilizer application, causing areas to be under-fertilized or over-fertilized. The design phase allows for the

development of mechanisms, like **pressure regulation valves**, to mitigate these risks and ensure a consistent flow of fertilizer.

Through careful analysis and modeling during the design phase, many of these risks can be anticipated and mitigated, ensuring the system performs reliably throughout its lifecycle.

1.2.3 Cost and Time Efficiency

While it might seem that the design phase is an investment in time and resources, it can actually save both **time** and **money** in the long run. A well-designed system can **prevent costly rework**, minimize errors during manufacturing, and ensure that the final product is functional from the first prototype. Let's explore how:

- **Prototyping and Testing**: During the design phase, concepts can be tested digitally through **computer-aided design (CAD)** simulations and **computational fluid dynamics (CFD)** modeling. These tools can help detect potential flaws, such as improper spray patterns, inefficient nozzle designs, or weak structural components, without needing to build physical prototypes. Catching these issues early on prevents the need for costly and time-consuming rework during later stages of production.
- **Material Selection**: Choosing the right materials is a crucial aspect of the design phase. Selecting high-quality materials that are both cost-effective and durable can result in **long-term savings** by reducing maintenance needs and improving the longevity of the drone. For instance, **high-density polyethylene (HDPE)** was chosen for the reservoir due to its low cost, light weight, and resistance to corrosion, which ensures that the drone performs well without the need for expensive or frequent repairs.
- **Manufacturing Efficiency**: A well-thought-out design helps streamline the manufacturing process. By considering factors such as ease of assembly, ease of component integration, and available production technologies (e.g., 3D printing, injection molding), the design phase ensures that the drone's parts can be manufactured quickly and at a lower cost.

In summary, the design phase enables the project to proceed with greater **efficiency**, reducing the risk of **costly mistakes** or **delays** that can occur later in the production process.

1.2.4 Facilitating Innovation and Optimization

The design phase is not just about following preset requirements; it is also about identifying opportunities for **innovation** and **optimization** that can enhance the drone's performance, reduce its environmental footprint, and improve the overall user experience. For example:

- **Flow Rate Optimization**: The nozzle system design could incorporate innovative features, such as **variable flow rate control**, which adjusts the spraying rate based on the drone's speed or GPS data. This ensures that the drone applies the right amount of fertilizer in the right place, reducing waste and improving accuracy.

- **Modular Design**: The reservoir and nozzle system could be designed with a **modular approach**, allowing for easy replacement or upgrades of components. If the drone's functionality needs to be enhanced in the future—perhaps by adding sensors to optimize fertilizer use—the modular design allows for easy adaptation without redesigning the entire system.
- **Energy Efficiency**: Another key aspect of optimization is **energy efficiency**. The design phase allows for the integration of **lightweight materials** and **aerodynamic features** to reduce power consumption during flight. This optimization can improve the drone's flight time, enabling it to cover larger areas without requiring frequent recharging.

By focusing on both optimization and innovation during the design phase, the drone system can not only meet current project goals but also accommodate future upgrades, thereby increasing the drone's value and sustainability over time.

1.2.5 Design for Sustainability and Environmental Impact

In today's world, **sustainability** is a driving factor in the design of many technologies, including agricultural drones. The design phase is where we can integrate **eco-friendly** practices that reduce the environmental impact of the drone during both its operation and its lifecycle. Here are some ways the design phase contributes to sustainability:

- **Material Selection for Longevity and Recycling**: Choosing materials that are both durable and recyclable helps reduce the environmental impact of the drone. HDPE, for example, is a highly recyclable material that can be reused at the end of the drone's lifespan. This reduces the need for new raw materials and minimizes waste.
- **Minimizing Fertilizer Wastage**: One of the primary goals of the fertilizer spraying drone is to **reduce chemical wastage**, which is a significant issue in traditional fertilizer application methods. The design of the nozzle system can contribute to this goal by optimizing the **spray pattern** and **flow rate**, ensuring that fertilizer is applied only where it is needed, minimizing the risk of over-application and reducing chemical runoff into the environment.
- **Energy Consumption Optimization**: The design phase also allows for the optimization of energy consumption. Lightweight and aerodynamic designs reduce the power required to fly the drone, making it more energy-efficient and reducing the overall carbon footprint.

Through thoughtful design choices, we can ensure that the drone is not only **economically viable** but also **environmentally responsible**, contributing to the long-term sustainability of agricultural practices.

1.2.6 Collaboration and Team Synergy

The design phase provides an opportunity for team members from different disciplines to collaborate, share ideas, and leverage their expertise. In our case, the drone project involved expertise in **aerodynamics**, **fluid dynamics**, **material science**, and **electronics**, among others. Each team member brought a unique perspective that enriched the design process:

- **Mechanical Engineers** contributed insights into structural integrity, payload limits, and aerodynamics.
- **Electrical Engineers** focused on how the reservoir and nozzle system could integrate with the drone's flight control system and sensors.
- **Chemical Engineers** helped with selecting materials resistant to fertilizers and ensuring safe, reliable flow rates.

This collaboration ensures that the final design integrates all aspects of the system seamlessly, allowing the drone to operate as an efficient and functional unit.

The **design phase** is undeniably one of the most critical stages in the development of any engineering project, especially when it comes to highly complex systems like the **automated fertilizer spraying drone**. By dedicating time and effort to understanding and addressing the technical, environmental, and practical constraints, we can create a design that not only meets the project's objectives but also lays a solid foundation for future success.

Through careful consideration of **performance requirements**, **risk mitigation**, **cost-effectiveness**, **innovation**, and **sustainability**, the design phase ensures that the drone system will function as intended while also being optimized for long-term

2. Design Requirements and Constraints

Designing the reservoir and nozzle system required a careful balance between multiple **technical**, **environmental**, and **economic** factors. These factors served as the basis for the design and helped us prioritize the most critical aspects of the system. Below is a detailed breakdown of the design requirements and constraints that influenced our design decisions.

2.1 Technical Requirements

- **Reservoir Capacity and Volume:** The primary requirement for the reservoir was that it had to be able to hold enough fertilizer to cover a specified area (typically up to 1 hectare). Based on typical application rates of fertilizers (e.g., **10 liters per hectare**), we estimated that the reservoir should have a capacity of **10 liters**. This figure ensures that the drone can cover a reasonably large area during one flight without requiring frequent refills. However, this volume also had to fit within the drone's **payload capacity**.
    - **Payload Limitations:** The drone could carry a total of **5-7 kg** of weight, including the weight of the reservoir, fertilizer, nozzles, electronics, and other components. A 10-liter fertilizer reservoir weighs about **10 kg** when filled

with liquid fertilizer, so we had to balance the need for capacity with the requirement for a lightweight design.

- **Spray Uniformity and Accuracy:** The nozzle system had to deliver a **uniform spray pattern** to ensure that the fertilizer was distributed evenly over the entire field. This uniformity was critical to avoid both under-application and over-application, which could lead to inefficient fertilizer use and potential harm to crops. We aimed to design a system that could maintain this uniformity despite varying flight speeds, heights, and wind conditions.
    - **Flow Rate:** The nozzle needed to maintain a consistent flow rate, regardless of changes in the drone's flight conditions, such as altitude or speed. This consistency would ensure that the amount of fertilizer dispensed was appropriate for the size of the field, reducing wastage and ensuring optimal crop growth.
- **Pressure Regulation:** The nozzle and reservoir system required a **pressure-regulation mechanism** to ensure the flow of fertilizer remained constant throughout the flight. As the drone's fuel consumption decreases and the reservoir empties, the pressure in the system fluctuates. A proper pressure regulation system would maintain constant flow rates despite these changes.
- **Material Selection:** Both the reservoir and the nozzle system needed to be made from **corrosion-resistant** materials that could withstand exposure to chemical fertilizers without degrading over time. Moreover, the materials had to be lightweight to minimize the total payload weight of the drone.

2.2 Environmental Constraints

- **Temperature Range:**
  The system was required to function in a variety of temperature conditions, from high temperatures in tropical climates to colder conditions in temperate regions. The materials used for the reservoir and nozzle had to withstand these fluctuations without degrading. For example, the fertilizer might freeze in colder climates, and we needed to ensure that the reservoir could handle the pressure changes that occur when liquids freeze.
- **UV Resistance:**
  The drone, along with its reservoir and nozzle system, would be exposed to prolonged sunlight during operations. Therefore, **UV-resistant materials** were chosen to prevent degradation from UV radiation, which can weaken plastic and rubber materials over time.
- **Chemical Resistance:**
  Fertilizers, particularly chemical-based fertilizers, can be highly corrosive. The reservoir and nozzle system needed to be designed with materials that would resist corrosion or degradation from the fertilizer chemicals, ensuring long-term durability.

2.3 Practical Constraints

- **Cost:**
  The design had to be cost-effective, especially since the drone was part of a larger agricultural automation project with a set budget. While high-end materials such as carbon fiber or titanium might be ideal in terms of strength and weight, we needed to balance performance with cost-effectiveness. Materials like **high-density polyethylene (HDPE)** for the reservoir and **brass or stainless steel** for the nozzle were chosen as they provide a good balance between durability, chemical resistance, and cost.
- **Manufacturing Feasibility:**
  We had to ensure that the components could be easily manufactured and assembled within the constraints of available resources. The reservoir was designed to be simple to fabricate with conventional manufacturing techniques such as **injection molding** or **rotational molding**, while the nozzle system required precise machining.

---

3. Design of the Reservoir

The **reservoir** is one of the most critical components of the drone, as it holds the fertilizer and ensures that the system can operate for a reasonable duration without needing to be refilled frequently. Below is a detailed explanation of the design and development process for the reservoir.

3.1 Initial Conceptualization

The initial brainstorming phase focused on determining the optimal size, shape, and material for the reservoir. Hari and I considered several possible configurations:

- **Cylindrical tank**: This was the first shape we settled on, as it offered an efficient use of space and a simple structural design. A cylindrical tank would also fit well under the drone without causing imbalance.
- **Spherical tank**: While this could potentially offer better strength-to-weight ratios, it was not ideal for mounting on the drone, as it would be harder to secure properly and would take up more space.

After further analysis, we decided that a **horizontal cylindrical design** would best meet the payload and volume requirements while providing an efficient shape for mounting on the drone.

3.2 Material Selection

We chose **high-density polyethylene (HDPE)** as the primary material for the reservoir. HDPE is widely used in agricultural and industrial applications due to its **high chemical resistance**, **low density**, and **cost-effectiveness**. Fertilizers, which often contain compounds like nitrogen and phosphorous, can be corrosive to many materials. HDPE's resistance to such chemicals made it an ideal choice for the reservoir, ensuring its longevity and durability.

Additionally, HDPE is **lightweight**, a key factor in keeping the drone's total weight within limits. The material was chosen in a **4mm thickness** to provide enough strength to withstand internal pressure and external forces while maintaining an optimal weight.

3.3 Reservoir Design Considerations

- **Volume and Capacity:**
  The reservoir was designed to hold **1 litres** of fertilizer, which was determined based on the required area coverage for a single flight. The choice of volume was just an estimation taken based on research done online for a testing flight.

3.4 Volume and Capacity:

The **reservoir** for the automated fertilizer spraying drone was designed to hold **10 liters of fertilizer**, a capacity chosen after careful consideration of several critical factors, including the drone's **payload limitations**, **flight duration**, and the specific needs of the agricultural application. Below, we explain the rationale behind choosing this volume, as well as the design trade-offs that influenced the final decision.

---

1. Area Coverage Requirement

One of the primary factors driving the design of the reservoir's volume was the **area coverage** required for a single flight. The purpose of the drone is to deliver fertilizer evenly across a designated area, typically a portion of a farm or an entire field. The amount of fertilizer needed per unit area is determined by a variety of factors, such as:

- **Type of fertilizer**: Fertilizers come in various types (liquid, granular, slow-release), each with different application rates. For liquid fertilizers, which are commonly used in aerial spraying, the application rate is usually given in liters per hectare (L/ha). The typical application rate for liquid fertilizers ranges between **10 to 20 liters per hectare**, depending on the crop and field conditions.
- **Field size**: The area to be covered in a single flight is another determinant. A large field requires a larger reservoir to ensure that the drone can cover it in one pass without needing to land and refill, which would reduce operational efficiency. Conversely, smaller fields might require smaller reservoirs, but for general farming operations, **10 liters** is a reasonable middle ground.

In this case, the reservoir's volume of **10 liters** was chosen because it can cover an area of approximately **1 hectare** (10,000 m^2) when applying **10 liters per hectare**. This ensures that the drone can perform a full round of spraying over a large area before needing to land and refill, maximizing efficiency.

---

2. Payload Limitations

The design of the reservoir was also heavily influenced by the **payload capacity** of the drone. The payload refers to the total weight that the drone can carry while still maintaining stable flight characteristics, such as **aerodynamics**, **lift**, and **control**. For this particular drone, the total payload capacity was set at approximately **5-7 kg**, which includes not only the **fertilizer** but also other components such as the **battery**, **sensor systems**, and **electronics**.

- **Weight of the Fertilizer**: Liquid fertilizers generally have a density of around **1.2 kg/L**, meaning that **10 liters** of fertilizer weighs approximately **12 kg**. Since the drone's payload capacity was limited to **5-7 kg**, we were faced with a challenge: the weight of the fertilizer alone exceeds the available payload.

Given that the fertilizer itself was too heavy for the current design constraints, we considered several options:

- **Reducing the reservoir size**: By reducing the volume of the reservoir to, say, **5 liters**, we could bring the weight down to **6-7 kg**. However, this would significantly reduce the area that the drone could cover in a single flight. In most large-scale agricultural applications, this trade-off would lead to inefficiencies, as the drone would need to land and refill more frequently.
- **Alternative liquid formulations**: Another option was to use a lighter fertilizer formulation, possibly with a lower density or even a slow-release liquid that requires a smaller volume per hectare. However, this would require additional research into different fertilizers that fit these criteria.

Ultimately, the decision was made to design a **composite, lightweight reservoir structure** and **optimize the drone's payload distribution**. This involved choosing lighter materials for the structure of the reservoir itself, such as **high-density polyethylene (HDPE)**, which is both lightweight and durable. The final design incorporated a **10-liter reservoir**, but the drone's other components (such as the battery and sensors) were designed to compensate for the weight of the fertilizer.

---

3. Flight Duration and Efficiency

The drone's ability to maintain **optimal flight time** while carrying the fertilizer was another key factor in determining the reservoir's volume. Drone flight time is influenced by several factors, including:

- **Battery life**: The capacity of the drone's battery must be sufficient to support the weight of the reservoir and fertilizer, along with the drone's operational power needs (flight control, GPS, sensors, etc.). The more weight the drone carries, the more power it requires, which in turn reduces flight time.

- **Fuel consumption**: The propulsion system, whether powered by batteries or an internal combustion engine, consumes more energy as the drone's weight increases. Since fertilizer is a liquid and the reservoir has to be large enough to hold a meaningful amount, any increase in weight leads to a direct decrease in the drone's operational efficiency in terms of fuel consumption.

For example, a **10-liter reservoir** will provide adequate coverage for approximately **1 hectare**, but the drone's flight time will be shorter when carrying this full load compared to flying with a lighter payload. The design goal, therefore, was to optimize flight duration through lightweight materials, efficient power systems, and aerodynamics, which would allow the drone to maximize its operational efficiency despite the weight of the fertilizer.

---

4. Design Trade-offs: Capacity vs. Practicality

Designing the ideal volume for the reservoir involves making key trade-offs between **capacity** and **practicality**. We could have opted for a larger reservoir, say **15 liters** or **20 litres**, which would have allowed the drone to cover larger areas before needing to land. However, this would have introduced several complications:

- **Increased weight**: A larger reservoir would increase the weight of the drone beyond the payload limits. This would require additional modifications to the drone's power system and may result in diminished **flight time** and **performance**.
- **Reduced manoeuvrability**: Carrying more weight could affect the drone's **manoeuvrability**, making it harder to control, especially in windier conditions or when flying at low altitudes. The increased weight could also require larger motors or a more powerful battery, both of which would add to the overall size and cost of the drone.
- **Structural integrity**: The drone's frame and propellers are designed to support a specific range of weights. Adding too much mass could jeopardize the **structural integrity** of the drone, potentially leading to failures or unsafe flight conditions.

Given these considerations, the design team concluded that **1 litres** was the optimal volume for the reservoir, as it provided a good balance between **coverage area**, **payload capacity**, and **flight performance**.

---

5. Optimizing the Fertilizer Distribution

The reservoir's design was also influenced by the need to ensure that fertilizer could be distributed evenly and efficiently during flight. The **nozzle system** must be able to handle the flow rate corresponding to the reservoir's volume without causing irregularities in the spraying pattern. A larger reservoir means a higher volume of fertilizer to be dispensed over a larger area, requiring careful engineering of the nozzle system to ensure:

- **Even distribution**: Regardless of the fertilizer volume, the spray pattern must remain consistent across the entire flight area. If the reservoir is too large or if the nozzle

system cannot handle the required flow rate, it may lead to uneven spraying or wastage.

- **Flow regulation**: The design of the reservoir also considered the need for **pressure regulation**. As the fertilizer is dispensed, the volume decreases, and pressure within the system may fluctuate. A well-designed **pressure control system** is essential to maintaining consistent flow throughout the entire spray cycle.

---

6. Future Considerations for Volume Adjustments

As drone technology evolves, there may be opportunities to adjust the reservoir's volume based on improvements in **battery efficiency**, **motor power**, or **aerodynamic design**. Future versions of the drone could potentially carry larger reservoirs, enabling them to cover larger areas without frequent refills. However, these adjustments will need to consider not only the **payload capacity** but also **energy consumption** and **operational efficiency**.

One approach could involve designing a **modular reservoir system**, where multiple smaller reservoirs could be added or removed depending on the specific requirements of the mission. This would provide greater flexibility, allowing the drone to adapt to different field sizes, crops, and fertilizer types.

In summary, the **1-liter capacity** for the reservoir was chosen based on a careful assessment of several factors, including the required area coverage, the drone's payload limitations, flight duration, and fertilizer distribution efficiency. This volume struck a balance between **operational efficiency** and the **physical constraints** of the drone, ensuring that the system could function effectively without overloading the drone or compromising its performance. The design of the reservoir was not just about maximizing capacity, but about integrating it into the drone's overall system design in a way that ensured reliability, cost-effectiveness, and sustainable operation.